

Agency:

- Villa, S., Barth, L. L., Chiossi, F., Welsch, R., and Kosch, T. (2025). Whose mind is it anyway? A systematic review and exploration on agency in cognitive augmentation. *Computers in Human Behavior: Artificial Humans*, 5, Article 100158.
- Bergström, J., Knibbe, J., Pohl, H., and Hornbæk, K. (2022). Sense of Agency and User Experience: Is There a Link? *ACM Transactions on Computer-Human Interaction*, 29(4), Article 28.

Sense of Community:

- McMillan, D. W., and Chavis, D. M. (1986). Sense of community: A definition and theory. *Journal of Community Psychology*, 14(1), 6–23.
- Chavis, D. M., Lee, K. S., and Acosta, J. D. (2008). The Sense of Community (SCI) Revised: The Reliability and Validity of the SCI-2. Paper presented at the 2nd International Community Psychology Conference, Lisboa, Portugal.

Since we have only four futures, they need to differ to be interesting for users and evoke discussion in the focus group, so we introduce interaction patterns. This is a design choice that does not affect the control the user has. A system can be interactive but low-agency if the AI controls the agenda, and a system can be less interactive but still preserve decision agency if the human retains control (Gomez et al. 2025).

- Gomez C, Cho SM, Ke S, Huang C-M and Unberath M (2025) Human-AI collaboration is not very collaborative yet: a taxonomy of interaction patterns in AI-assisted decision making from a systematic review. *Front. Comput. Sci.* 6:1521066. doi: 10.3389/fcomp.2024.1521066
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Seven Interaction patterns of AI from Gomez et al. are:

1. AI-first - AI decision shown simultaneously with the problem; user had no prior independent assessment
2. AI-follow - User forms own assessment first, then AI's outcome is shown for comparison
3. Secondary assistance - AI offers supplementary context; user interprets relevance themselves -
4. Request-driven - User actively asks for AI input; AI doesn't proactively present
5. Dialogic - AI asks questions, collects user context iteratively, then proposes outcome
6. User-guided interactive adjustments - User modifies inputs to reshape AI's outcome
7. Delegation - One agent (user or collective) hands decision authority to the other

We can turn that into->

High agency: **Request-driven AI assistance**; User-guided interactive adjustments; human-controlled Delegation; Secondary assistance and **AI-follow assistance**.

Low agency: **AI-first assistance; AI-controlled Delegation**; some AI-guided dialogic systems where the AI tightly scripts the exchange.

Future A: Collective negotiation
High agency × high community
Pattern: request-driven/dialogic

Future B: Individual, AI second opinion
High agency × low community
Pattern: AI-follow/request driven

Future C: Community delegation
Low agency × high community
Pattern: delegation

Future D: Automated allocation
Low agency × low community
Pattern: AI-first

Future A (dialogic, high agency × high community). AI-guided dialogic user engagement pattern: "the AI facilitates a dialogue-like engagement with the user. Guided by the AI's instructions, the user responds by providing pertinent information. The iterative exchange continues until the AI's instruction requirements are satisfied." The participant enters their preference, sees aggregate building preferences and the collective constraint, and the AI facilitates a back-and-forth that surfaces trade-offs explicitly (for example, "If you take 22°C, the household on floor 3 with an elderly resident gets 27°C. Would you like to adjust, or see other arrangements?" or suggest sharing electricity with neighbors). The AI proposes balanced allocations and prompts the user to confirm or revise.

Future B (AI-follow, high agency × low community). AI-follow assistance pattern: "user forming an independent preliminary prediction. Following this initial judgment, the AI's predicted outcome is presented." The participant enters a preferred temperature first. The AI then responds explicitly with a constrained counter-offer, a stated rationale, and one or two alternatives. The participant accepts, modifies, or rejects. The AI is concrete and visible as an actor.

Future C (delegation, low agency × high community). Delegation pattern: "both the user and the AI leverage their unique strengths and capabilities to optimize the decision-making outcome." The interface/ARKI has pre-delegated heatwave allocation to a collective rule operationalised by ARKI The participant sees their share, the collective rule, and the building's distribution. To change requires invoking a community-override process framed as a request for exception.

Future D (AI-first, low agency × low community). AI-first assistance pattern:

"decision-making problem and the AI-predicted outcome are simultaneously displayed to the user." The interface opens with the ARKI allocation already in place, accompanied by a brief grid-condition rationale. The participant sees only their own allocation. To change it, they must explicitly invoke an adjustment action framed as a deviation from the system's plan. Inaction means accepting the AI's allocation.

Axis definitions and mapped focus group questions

Axis 1: Agency

Core definition (Villa et al., 2025, p. 3, Section 1.1.3): "The sense of agency refers to the experience of controlling one's actions and their consequences (Limerick et al., 2014; Moore, 2016), allowing us to recognize ourselves as the agent of our behavior (Jeannerod, 2003)."

- **Two-level distinction (Bergström et al., 2022, p. 28:3, Section 2):** "a distinction is drawn between two levels of agency: a low-level, implicit sense of control, and a higher-level, explicit experience of control." The "I did that" feeling and the experience of "being the author or controller of one's actions" sit at the explicit level and are what self-report questions can reach.
- **AI-specific operationalisation (Villa et al., 2025, p. 3, Definition #2):** "Agency in human augmentation is an individual's sense of control and purposeful engagement with technological tools that enhance cognitive abilities. It means actively choosing, directing, and managing technological interventions that support personal cognitive functions like memory and decision-making, ensuring the person remains the primary driver of their enhanced capabilities (Bennett et al., 2023)."
- **Assistance-agency trade-off (Coyle et al., 2012, summarised in Villa et al., 2025, Section 2.3.1):** "computer assistance up to a certain level can still allow a high sense of agency. However, medium and high assistance significantly reduced agency." Agency does not scale linearly with assistance; there is a threshold beyond which it collapses, and finding that threshold is a design question. **This is important!**
- **Outcome bias on agency (Tajima et al., 2022, summarised in Villa et al., 2025, Section 2.3.1):** "participants reported a higher sense of agency for favorable outcomes (assistive-touch) as compared to unfavorable outcomes (adversarial-touch). This suggests that the level of perceived agency is affected by an outcome bias." This is directly relevant to the study's unfavorable-outcome manipulation, since outcome valence can confound the agency reading.

High agency:

Gomez et al. mapping: Primarily AI-follow assistance because the participant forms an independent preliminary plan before seeing AI advice (Gomez et al., 2025, p. 5). The “Check Plan with ARKI” action adds request-driven AI assistance, where the user controls when assistance appears. Live energy feedback is secondary assistance because it informs the decision without directly deciding the temperatures (p. 5). If ARKI updates its recommendations after user edits, the prototype also includes user-guided interactive adjustments

- **Visual hierarchy:** user controls visually dominant, min. 60% of view (Coyle et al., 2012).

Low::

Gomez et al. mapping: This is primarily AI-first assistance because the AI-produced outcome is visible before the participant forms an independent plan (Gomez et al., 2025, p. 5). The exception request is a limited request-driven interaction, but it does not move the cell into high agency because ARKI's plan remains operative while the request is reviewed.

HIGH LOW STILL edits to be made

Axis 2: Sense of community: Personal frame ↔ Shared frame

Definition

Sense of others is the degree to which the interface frames the user's consumption as drawing from a finite shared resource that others also depend on, rather than as a private transaction between the user and the service. It draws on the SusAF social dimension (Sense of Community) and environmental (Betz et al., 2024), specifically the topics of Energy and Biodiversity and Land Use, which ask whether an IT product makes the ecological consequences of consumption visible to those who cause them. It draws directly on the Nature-Positive Lifestyles report (Galli et al., 2026), which argues that lifestyle impacts on biodiversity have remained "largely invisible to decision-makers" because consumption interfaces frame the act as private, and that making the shared and finite character of the resource visible is a precondition for behavioural change. It also connects to McMillan and Chavis (1986) who describe a sense of community from four parts: membership (belonging), influence (mattering to the group), integration of needs (the group meets your needs and you meet theirs), and shared emotional connection.

Eco-feedback (Froehlich et al., 2010 CHI) define eco-feedback as feedback that "provides feedback about individual or collective behaviour and aims to persuade people to improve their

environmental footprint." **here, the eco-feedback framing is available in both cells in detail, but the framing depends if it's high/low.**

McMillan, D. W., & Chavis, D. M. (1986). Sense of community: A definition and theory. *Journal of Community Psychology*, 14(1), 6–23.

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The research question behind this axis (drawn from SusAF): How can the interface have an effect on a person's sense of belonging to different groups?

High Sense of others

The interface makes others visible. The user can see, in the interface itself, that their consumption draws from a finite resource shared with named others - building, neighbourhood, city, food, and that what they take changes what remains for the rest.

- **Shared identity.** The interface connects users to a group, place, or collective purpose. Example: "Group used X this week" The interface states shared things and how many parties draw from it. The group's collective impact is documented and archived (cumulative biodiversity footprint reduction, shared milestones, crises resolved together). McMillan & Chavis (1986): shared emotional connection requires identification with the group's history; the more important the shared event, the greater the bond. Nature-Positive Lifestyles report: nearly 70% of climate impacts are linked to household consumption; collective tracking makes the group's shared contribution to that figure visible and therefore emotionally meaningful.
- **Members are publicly honored for contributions.** McMillan & Chavis (1986): reward in the presence of the community is a direct cohesion mechanism. Hansson et al. (2021, citing Petkov et al., 2011): community-based comparative feedback triggers motivation in ways individual feedback cannot.
- **Visibility.** Other users are present in the interface, by name, count, or aggregate. The user can see they are not alone. The user's allocation is shown alongside the pool total, not separately. The visual relationship between the user part and the whole is shown.
- **Relational consequences.** The user's actions are framed in terms of what they do for or with others. Example: "If you and X% of other group members join, you meet the goal of Y" The interface surfaces the fact that other parties' consumption affects what is available to this user.
- **Influence:** members can propose, vote on, or shape the group's next collective nature-positive target. McMillan & Chavis (1986): members are more attracted to communities where they feel influential; voluntary conformity is stronger than imposed conformity. Nature-Positive Lifestyles report: choice-editing works best when communities redesign their own consumption defaults rather than receiving them top-down.
- **Shared norms and outcomes.** The interface surfaces group-level results and collective goals. The norms are based on the idea that norms are changed towards better.

- **Mutual aid.** Helping or being helped by others is part of the interaction, not a side feature. Example: a household with extra resources can give it to a neighbor. A mechanism exists for unused X to flow back to the pool, and using it is part of the main interaction. Example: "You have remaining X, want to give it away to Y?"
- **Mutual dependence.** Users feel that others depend on them, or that they depend on others, through the interface. They see their collective histories and their shared consumption.

<https://dl.acm.org/doi/10.1145/1753326.1753629>

Low Sense of others

The interface frames the user's consumption as a private matter between the user and the service. Whatever shared limits exist are not surfaced. The user sees their own number, their own history, their own consequences.

- **No shared identity.** The interface speaks only to "you." There is no "we," no city, no others, no peers. no archive of past collective actions or shared milestones; events end with no closure or group summary; no public honoring of members; outcomes attributed to the system or platform rather than to the group. McMillan & Chavis (1986): shared emotional connection requires a shared story; closure is required for cohesiveness. Nature-Positive Lifestyles report: lifestyles are "socially shaped patterns reflecting group dynamics"; without a group story, individual consumption behavior remains tethered to the dominant system rather than to the community.
- **Invisibility.** Other users do not appear in the interface, even in aggregate. The user has no idea anyone else is using the interface.
- **Individual consequences only.** Outcomes are framed as users own. No collective stakes are surfaced. The ecological equivalences shown in the baseline are individualised to this user's consumption
- **No shared norms or outcomes.** No peer data, no group goals, no collective metrics. Nothing in the interface indicates that other people's choices affect what is available to this user. If any limits exist in the interface, they are framed only for the one user, not as collective goals.
- **No mutual aid.** Helping or being helped by others is not part of the design. The interface optimizes for one user at a time.
- **No interdependence.** Nothing in the interface suggests that others rely on this user or that this user relies on others. Nothing in the interface indicates that other people's choices affect what is available to this user. If any limits exist in the interface, they are framed finitely.
- **Influence:** the system or platform sets all sustainability targets; no voting, proposing, or co-authoring of group norms; contributions appear without author attribution. McMillan & Chavis (1986): the force toward uniformity is transactional; removing member input

eliminates one side of it. Hansson et al. (2021): critique of persuasive eco-feedback systems centers on exactly this; the assumption that individuals will act when given top-down information is the "limited framing" the literature repeatedly flags.

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